Polymarket Relative Value Terminal

> A relative-value terminal for prediction markets. Scans structurally linked market families for dislocations that violate logical constraints, visualizes the opportunity, and builds the spread.

Pitch

We turned Polymarket into something traders can reason about like an options desk. Same underlying, different strikes, different expiries — and we show you when the surface is broken.

The Pivot

Previous versions built a recommendation engine around hand-tuned weights connecting loosely related macro markets (Fed → BTC). The relationship was always the weak point — heuristic, attackable, and dependent on narrative.

New approach: use Polymarket's own market structure. The relationships are definitional, not estimated:

Family type	Constraint	Example
Strike ladder	Monotonic: P(above $100k) ≤ P(above $90k) ≤ P(above $80k)	BTC price threshold markets
Expiry curve	Term structure: near-expiry ≤ far-expiry for "by date X" markets (monotonic in time)	"Fed holds by May" / "by June" / "by July"
Mutually exclusive set	Sum to ~100%: P(A) + P(B) + P(C) ≈ 1.00	"Who wins?" outcome markets
Nested events	Implication: P(specific) ≤ P(general)	"Lakers win finals" ≤ "Lakers make finals"

No heuristic weight. The constraint IS the market structure. A violation is a dislocation by definition, not by estimation.

Core Math

Dislocation Detection

Strike ladder:

```
For markets M_1 (above $X_1), M_2 (above $X_2) where X_1 < X_
2:
  constraint: price(M_1) ≥ price(M_2)
  dislocation = price(M_2) - price(M_1)  when positive
```

Expiry curve:

```
For "by May" (M_near) and "by July" (M_far):
  constraint: price(M_far) ≥ price(M_near)
  dislocation = price(M_near) - price(M_far)  when positive
```

Mutually exclusive:

```
For outcomes {M_1, ..., M_n}:
  constraint: Σ price(M_i) ≈ 1.00
  overpricing = Σ price(M_i) - 1.00
```

Nested:

```
P(specific) ≤ P(general)
dislocation = price(specific) - price(general)  when positive
```

Dislocation Severity

```
severity = |dislocation| / historical_std_of_dislocation(trai
ling 30d, PIT)
```

```
MINIMUM HISTORY: ≥20 prior daily bars. Suppress if fewer.
```

No percentile-based triggering on a heuristic gap. Severity measured against the pair's own structural spread history.

Token Economics (unchanged)

```
cost = token_price × shares
max_loss = cost
max_gain = (1 - token_price) × shares
NO price = 1 - YES price
```

Sizing — Equal Dollar-at-Risk

```
shares_i = target_risk_per_leg / token_price_i
```

Why: binary tokens have no continuous vol. Equal-risk is the simplest defensible baseline. Stated in UI.

EV

```
EV_leg    = shares × (my_prob_wins - token_price)
EV_pair   = Σ EV_leg - friction
EV_per_$  = EV_pair / total_max_loss
```

All numbers from calc module. No hand math.

Product — Three Screens

Screen 1: Universe Scanner

The product scans active Polymarket families and ranks dislocations by severity. This is what loads on launch.

```
┌──────────────────────────────────────────────────────────────┐
│  DISLOCATION SCANNER                        Live · 3 families │
├────────────────┬────────────┬──────────────┬────────────────┤
│  Family        │  Type      │  Severity    │  Status        │
├────────────────┼────────────┼──────────────┼────────────────┤
│  BTC strikes   │  Ladder    │  2.4σ        │  Actionable    │
│  Fed expiry    │  Curve     │  1.8σ        │  Watchlist     │
│  GOP primary   │  Mutex     │  0.6σ        │  Normal        │
└────────────────┴────────────┴──────────────┴────────────────┘
```

Actionable (≥2σ), **Watchlist** (1.5–2σ), **Normal** (<1.5σ).

This removes the cherry-picking problem. Judges see a systematic scan, not a pre-selected pair.

API: `GET gamma-api.polymarket.com/markets` + `GET gamma-api.polymarket.com/events` (event grouping surfaces families automatically)

Screen 2: Dislocation Visualizer (the hero visual)

Click a family → the visual opens. This is the centerpiece of the demo.

For strike ladders: Probability vs. strike chart. Markets plotted as points. Monotonic constraint shown as a shaded envelope. Violations are visually obvious — a point above the one to its left.

For expiry curves: Probability vs. expiry date. Term structure curve. Inversions highlighted.

For mutually exclusive sets: Stacked bar or pie showing sum vs. 1.00. Overpricing shown as the excess segment.

For nested events: Parent-child tree with probability bars. Violation = child bar exceeding parent.

Each visualization shows:

- Current prices (live from Gamma)
- 30-day history of the dislocation spread (from CLOB prices-history)
- Severity score (σ from structural spread history)

- Constraint explanation: one sentence stating why this relationship must hold

This is the visual judges remember. Not a ticket. Not a paragraph. A surface that makes a broken market structure instantly visible.

Screen 3: Spread Builder + Evidence

Once the dislocation is visible, the user builds the corrective spread.

Auto-generated spread: The direction is structurally implied — no heuristic, no convergence mode analysis needed:

Dislocation type	Spread structure
Strike violation: P($100k) > P($90k)	Buy NO on $100k, Buy YES on $90k
Expiry inversion: P(May) > P(July)	Buy NO on May, Buy YES on July
Mutex overpricing: sum > 1.00	Sell (buy NO) the most overpriced outcome
Nested violation: P(child) > P(parent)	Buy NO on child, Buy YES on parent

The direction is logically forced. No decision table needed. No convergence mode. The constraint tells you exactly which side is wrong.

Evidence panel:

Episode-based, point-in-time, same framework as before but cleaner:

```
TRIGGER:  dislocation severity crosses above 2σ (PIT, ≥20 bar
s trailing)
CLOSED:   dislocation returns to within 0.5σ within 7 days
FAILED:   still above 0.5σ after 7 days

Episodes:    N
Closed:      X/N
Median time: Y days
Worst loss:  Z per $100 risked (conservative friction proxy)
```

Evidence confidence: ≥10 episodes High, 5–9 Moderate, <5 → downgrade to "Structural anomaly — watchlist only, no spread recommended."

Backtest (point-in-time, leave-one-out):

```
For each episode e_i:
  1. Compute trigger severity using only trailing data (PIT)
  2. Enter auto-generated spread (direction from constraint)
  3. Equal-dollar-at-risk sizing
  4. Entry friction: conservative proxy (current spreads)
  5. Exit on ≤0.5σ or day 7
  6. Exit friction: spread × 1.2
  7. Record net P&L

Three friction scenarios: light (0.8×) / base (1.0×) / harsh
(1.5×)
```

Output:

```
─── BACKTEST (PIT, leave-one-out) ─────────
Episodes:     N
Wins:         W/N
Median P&L:   +$X / $100 (base friction)
25th pctl:    -$X / $100
Worst loss:   -$X / $100
Friction:     conservative proxy (current spreads)
⚠ Small sample. Past ≠ future.
───────────────────────────────────────────
```

Spread ticket:

```
──────────────────────────────────────────
SPREAD TICKET
──────────────────────────────────────────
Family:    BTC Strike Ladder
Violation: P($100k) = 0.42 > P($90k) = 0.38

Leg A: YES BTC >$90k   [shares] @ $[price]
Leg B: NO  BTC >$100k  [shares] @ $[price]
```

```
Net cost / Max loss:  [computed]
Max gain:             [computed]
EV/$ risked:          [computed]

─── Constraint ──────────────────────────
P(above higher strike) must be ≤ P(above lower strike).
Current violation: +4pp. This is structural, not estimated.

─── Evidence ────────────────────────────
Severity: 2.4σ (PIT, 30d trailing)
Episodes: N ([confidence])
Closed:   X/N, median Y days
Backtest: W/N wins, median +$Z/100 (base friction)
Worst:    -$Z/100

─── Model Risk ──────────────────────────
Friction is proxy (current spreads, not historical).
Small sample. Structural constraint is definitional.

─── Execution ───────────────────────────
Leg A → polymarket.com/event/[slug]
Leg B → polymarket.com/event/[slug]
────────────────────────────────────────
```

No-Trade Gates

Suppress spread recommendation if ANY is true:

```
1. Evidence confidence < Moderate (<5 episodes)
2. Median backtest P&L < 0 under base friction
3. 25th pctl P&L < -25% of risk under base friction
4. Severity < 1.5σ (normal range)
5. Signal collapses under harsh friction
6. Insufficient history (<20 trailing bars)
```

Output: **"Structural anomaly detected. Insufficient evidence for spread. Watchlist only."**

"Most tools force a trade. Ours explicitly refuses when validation fails."

Market Families — Systematic, Not Cherry-Picked

Discovery Method

1. Pull all active markets from Gamma API
2. Group by event (Gamma `events` endpoint provides this)
3. Classify family type by parsing market titles:
    - Numeric thresholds in same event → strike ladder
    - Date-varying markets for same outcome → expiry curve
    - Multiple outcomes under one question → mutually exclusive
    - Specific/general pairs → nested
4. Compute dislocation severity for all families
5. Rank and display

Pre-Verified Seed Families (fallback if parsing is slow)

Family	Type	Markets	Why clean
BTC price thresholds	Strike ladder	>$80k, >$90k, >$100k	Monotonic by construction
Fed rate hold by month	Expiry curve	by May, by June, by July	Term structure by construction
[Political outcome set]	Mutually exclusive	Multiple named candidates	Must sum to ~1.00

Emergency 4th family prepared privately. All verified live + liquid in Hour 0–1.

Hero by Evidence

Same gate + tie-break system:

Gates: ≥5 episodes, execution quality ≥ Moderate, median backtest ≥ 0 (base), severity ≥ 2σ currently.

Tie-break: most episodes → lowest worst loss → highest severity.

Kill switch: no family passes → signal-analysis mode. Show scanner + visualizer. Suppress spread builder. Frame as strength.

Hero locked before demo.

API Failure Strategy

Cached snapshots. Hardcoded seeds for 3 families. Green/amber/red badges. Never breaks.

Endpoints

Endpoint	Purpose	Confirmed
`GET gamma-api.polymarket.com/markets`	Prices, volume, spread	Yes
`GET gamma-api.polymarket.com/events`	Family grouping	Yes
`GET clob.polymarket.com/prices-history`	History, evidence, backtest	Yes
`GET clob.polymarket.com/book`	Bid/ask	Yes

Build Plan

Phase 1 (6–7 hours) — the submission

Hour 0–1: Family discovery + hero selection. Pull markets. Group into families. Classify types. Verify 3+ families live and liquid. Compute episodes + backtest. Apply gates. Lock hero.

Hour 1–2: Calc module. Token math, sizing, EV, dislocation severity, episode detection, backtest loop. Snapshot-test all paths.

Hour 2–4: Scanner + visualizer. Universe scanner table (Screen 1). Hero visual: probability vs. strike/expiry chart with constraint envelope and violation highlighting (Screen 2). 30-day dislocation history. Severity display. API integration + fallback.

Hour 4–5.5: Evidence + spread builder. Episode evidence panel. Confidence labels. PIT backtest with 3 friction scenarios. Auto-generated spread with structurally implied direction. Spread ticket from calc module.

Hour 5.5–6.5: Polish + no-trade. No-trade output. Watchlist mode. Loading/error states.

Hour 6.5–7: Hostile QA. Test all critical paths:

1. Strike violation → correct spread direction
2. Expiry inversion → correct spread direction
3. Mutex overpricing → correct sell target
4. Normal (no dislocation) → watchlist only
5. Insufficient evidence → suppression
6. Negative backtest → suppression
7. Insufficient history → trigger suppression
8. One-leg spread ticket → correct math
9. Two-leg spread ticket → correct math
10. API failure → fallback displays correctly

No demo until all 10 pass.

Sacrifice Order

1. Nested-event family type (keep strike + expiry + mutex)
2. Replay animations (static best/worst panels enough)
3. Robustness tables
4. **Never cut:** scanner, hero visual, evidence, correct spread logic, ticket

Phase 2 (3–4 hours)

Full replay. Additional family types. Scenario exit analysis.

Phase 3 (2 hours)

WebSocket live updates. Export. Broader universe.

What Changed from v9

Before (v9)	After (v10)	Why
Cross-domain macro pairs	Structurally linked families	Relationship is definitional, not estimated
Heuristic weight	No weight needed	Constraint IS the structure
Convergence mode decomposition	Not needed	Direction implied by constraint violation
Complex decision table	Simple: constraint tells you which side is wrong	No judgment calls possible
Ticket as hero	Dislocation visualizer as hero	Visual impact for judges
3 pre-selected pairs	Systematic scanner → drill into best	Removes cherry-picking
"Pair Analyzer"	"Relative Value Terminal"	Stronger product thesis

Disclaimer

Informational/educational only. Not financial advice. Structural constraints are definitional but market microstructure (fees, slippage, timing) affects realized P&L. Backtest uses PIT triggers, LOO validation, conservative friction proxy. Small samples. Past ≠ future. Tool suppresses recommendations when evidence is insufficient.